Predicting UFC Fight Outcomes Using Pre-Fight Fighter Statistics

Compulsory Exercise 2 — TMA4268 Statistical Learning V2026

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Abstract

We address the problem of predicting the winner of a UFC mixed martial arts fight from pre-fight statistics only. The target is a binary outcome: whether the fighter assigned to the Red corner wins (`WinnerRed` = 1) or loses (`WinnerRed` = 0). The dataset is the *Ultimate UFC Dataset* sourced from Kaggle (`mdabbert/ultimate-ufc-dataset`), containing 6,528 bouts between March 2010 and December 2024 with per-fighter cumulative statistics recorded at fight time.

We construct *difference features* — each numeric statistic is replaced by its Red-minus-Blue signed difference — to make the feature space matchup-symmetric and reduce dimensionality from 118 to 26 predictors. Three model families are compared: penalised logistic regression (Elastic Net, tuned with nested cross-validation), a Random Forest ensemble, and gradient-boosted trees (XGBoost). All models are evaluated on a chronological 80/20 holdout split to prevent time leakage, and benchmarked against the bookmaker-implied win probability derived from pre-fight betting odds.

The bookmaker baseline achieves an accuracy of 69.6% and a ROC AUC of 0.743. Our best-performing model, the Random Forest, reaches an accuracy of 63.6% and a ROC AUC of 0.693. The gap to the bookmaker confirms that publicly available fighter statistics contain real predictive signal but fall well short of the full information encoded in market odds. Among observable predictors, average takedown-landed difference, reach advantage, and win-streak difference emerge as the strongest predictors of fight outcome, while a negative age coefficient reveals that older Red-corner fighters tend to underperform relative to the prediction.

1 Introduction

Mixed martial arts (MMA) is one of the fastest-growing combat sports globally, and the UFC is its premier organisation. Each fight is a highly complex event influenced by physical attributes, technical skills, recent form, and matchup dynamics. Predicting fight outcomes is a canonical classification problem: given observable pre-fight information about both fighters, can we assign a probability that the Red-corner fighter wins?

Dataset. We use the *Ultimate UFC Dataset* published by user `mdabbert` on Kaggle (`mdabbert/ultimate-ufc-dataset`). The dataset records 6,528 bouts with 118 raw columns, including per-fighter cumulative career statistics (wins, losses, streaks, striking and grappling averages, physical measurements), contextual fight information (weight class, number of rounds, gender), and bookmaker moneyline odds. After filtering to bouts with a decisive Red or Blue winner and dropping pre-computed difference columns provided by the source, we retain 6,275 bouts for analysis.

Task and scope. Our goal is to build a classifier that, given only pre-fight information, assigns a calibrated probability of a Red-corner victory. We benchmark every model against bookmaker-implied probabilities — a near-market-efficient baseline that absorbs information beyond public statistics. The intended audience is a sports analytics practitioner who wants to understand which statistical learning method extracts the most signal from observable fighter data, and which pre-fight features drive predictions.

Practical constraint. Because betting-odds leakage would trivially dominate any model, odds-derived columns are excluded from all feature sets; the odds appear only as a benchmark in evaluation.

2 Descriptive Data Analysis

2.1 Dataset overview

After cleaning, the dataset contains 6,275 bouts spanning March 2010 to December 2024. The outcome is moderately imbalanced: 58.1% of bouts result in a Red-corner victory, as shown in Figure 1. This imbalance reflects the UFC's convention of assigning the Red corner to champions and higher-ranked fighters, who naturally win more often. Table 1 summarises the key numeric predictors used in modelling.

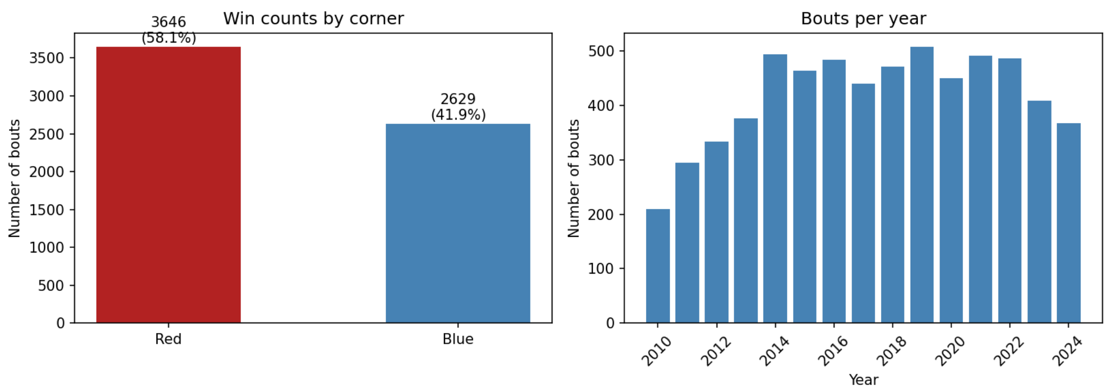

Figure 1: Left: Red vs Blue win counts in the cleaned dataset (6,275 bouts). Red-corner fighters win 58.1% of bouts. Right: number of bouts per year; data coverage is near-complete from 2013 onwards.

Table 1: Summary statistics for selected difference features (*Red minus Blue*) on the full cleaned dataset ($n = 6{,}275$). Missing-value counts differ per feature because career statistics were not tracked for early bouts.

Feature	Mean	Std	Min	Max
`AgeDiff`	0.56	5.18	−16.0	17.0
`ReachCmsDiff`	0.23	8.62	−30.5	188.0
`CurrentWinStreakDiff`	0.14	1.87	−10.0	18.0
`CurrentLoseStreakDiff`	0.12	1.01	−5.0	6.0
`AvgSigStrLandedDiff`	0.32	17.4	−128.2	104.7
`AvgSigStrPctDiff`	0.01	0.14	−0.57	0.82
`AvgTDLandedDiff`	0.09	1.79	−10.9	11.0
`WinsDiff`	1.49	4.17	−23.0	28.0

The near-zero means for the difference features are expected: the UFC broadly matches fighters of comparable ability, so the average edge is close to zero. The positive mean for `WinsDiff` (1.49) and `AgeDiff` (0.56) reflects the Red-corner assignment policy: Red fighters tend to have slightly more career wins and are slightly older.

2.2 Missingness

Many career-statistics columns have substantial missing values for early bouts where tracking was incomplete. Figure 2 shows the completeness rate per column. The *diff* feature set strategy (Section 3) retains all rows and delegates imputation to the model pipeline.

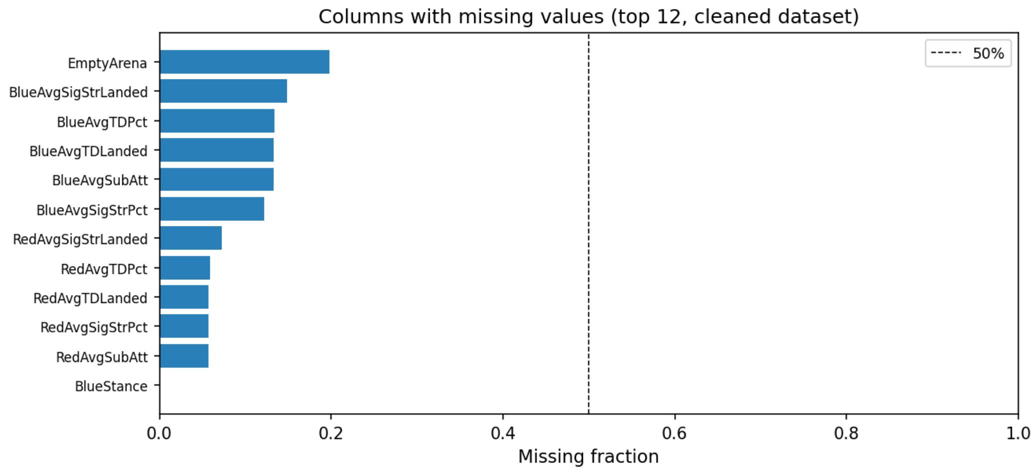

Figure 2: Fraction of non-missing values per column in the raw dataset. Fighter striking and grappling averages have notably lower completeness for older bouts.

2.3 Correlation with outcome

Figure 3 shows the Pearson correlation of each numeric raw column with the binary outcome `WinnerRed`. Correlations are modest ($|r| < 0.12$), confirming that no single feature is strongly predictive on its own. Win streak, cumulative wins, and significant-strike statistics show the highest marginal associations.

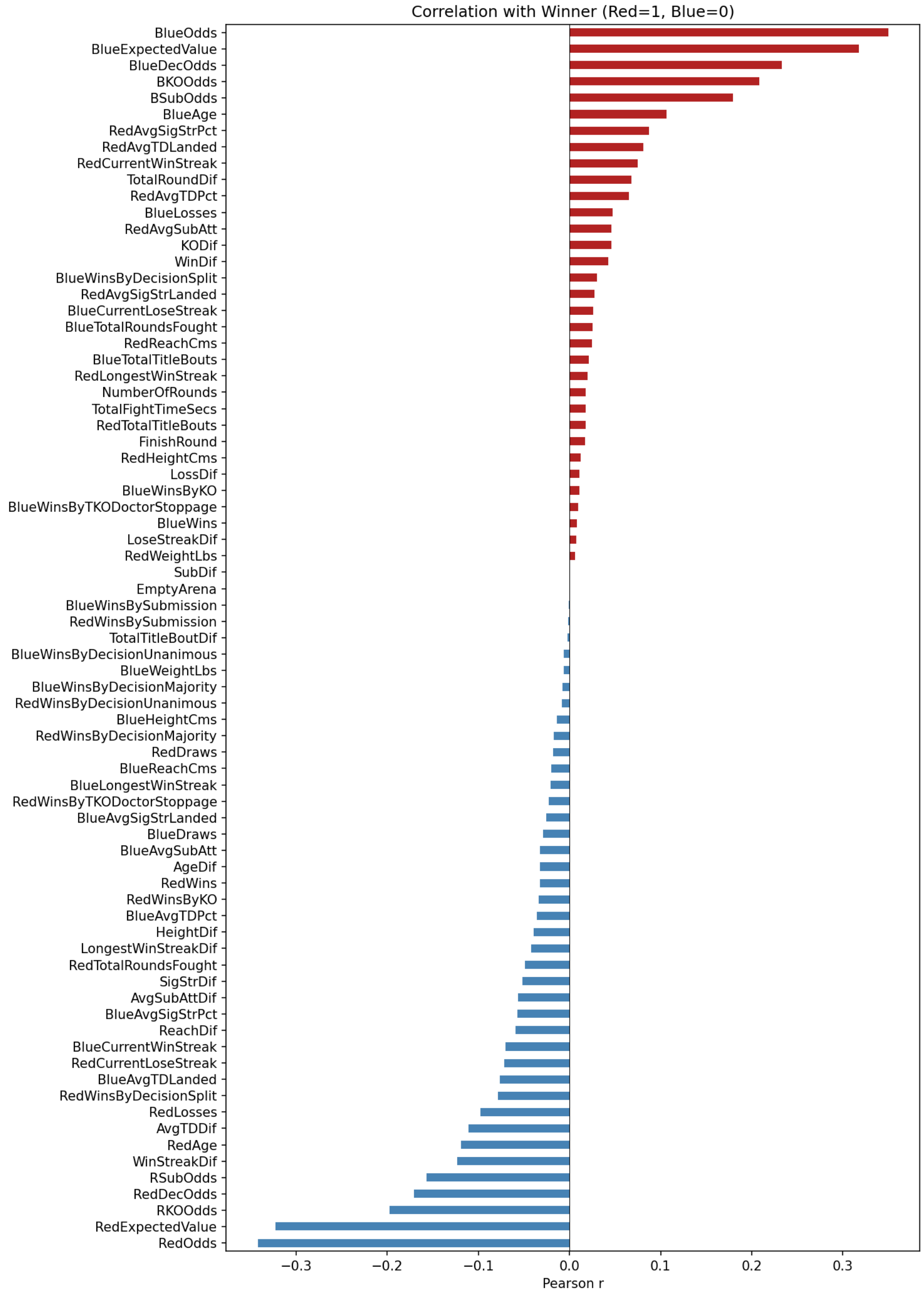

Figure 3: Pearson correlation of raw numeric columns with `WinnerRed`. The strongest individual predictors are win-streak and total-wins differences (Red corner), all with $|r| < 0.12$.

2.4 Feature distributions by outcome

Figure 4 shows violin plots of the six most important difference features split by fight outcome. The distributions largely overlap, which is consistent with the low individual correlations. The signal is subtle: in Red-win fights the win-streak and reach distributions shift slightly upward, confirming a real but weak effect.

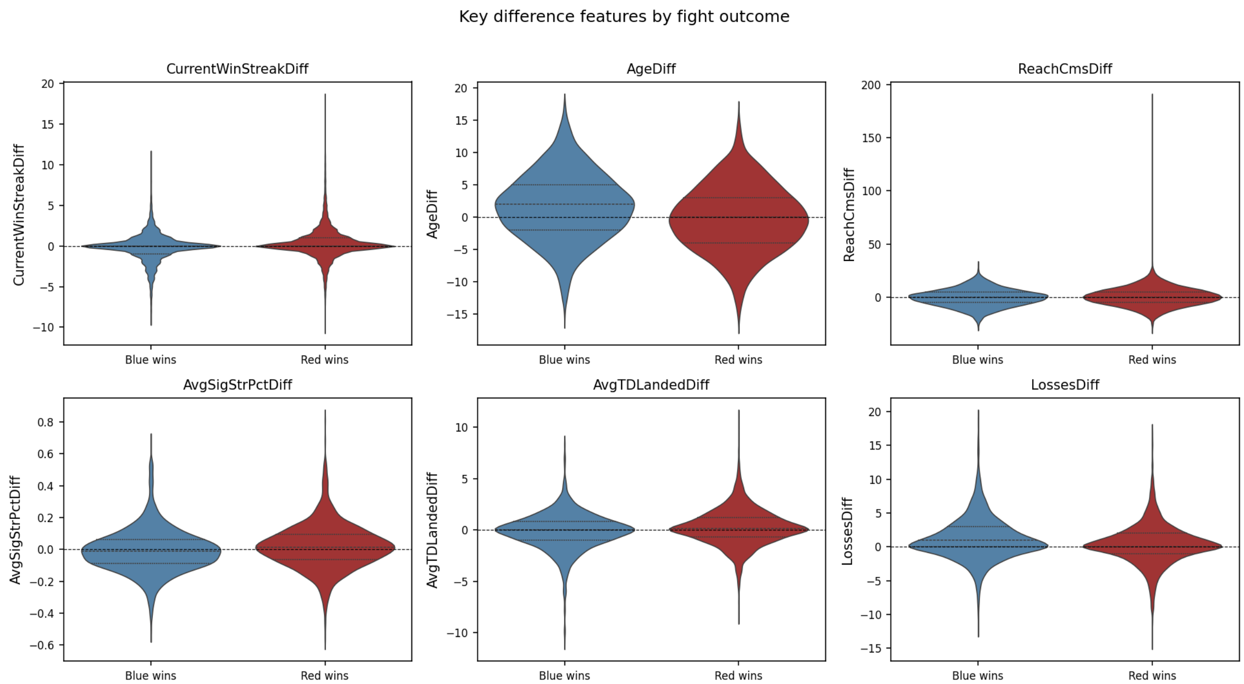

Figure 4: Distribution of key difference features (Red minus Blue) split by outcome. Dashed lines mark zero (even matchup). Distributions overlap substantially, reflecting the difficulty of the prediction task.

2.5 Feature correlation structure

Red and Blue fighter statistics are highly internally correlated (e.g., wins is correlated with win streak and total rounds fought). This motivates using difference features to compress correlated Red/Blue column pairs into a single signed quantity, reducing multicollinearity in linear models.

3 Methods

3.1 Feature engineering: the difference representation

The raw dataset contains symmetrically paired Red/Blue columns for every fighter statistic. We replace each pair $(x_{\text{red}}, x_{\text{blue}})$ with the signed difference $\Delta x = x_{\text{red}} - x_{\text{blue}}$. This yields a matchup-symmetric feature space: swapping Red and Blue negates all difference features and flips the outcome, so the representation is not corner-biased. The resulting *diff* feature set contains 26 predictors (numeric differences plus categorical context variables: gender, title-bout indicator, and number of scheduled rounds).

Missing values are imputed inside the model pipeline using column medians for numeric features and mode for categorical features, so all 6,275 bouts are preserved.

3.2 Data split strategy

All models use a **chronological holdout split**: fights are sorted by date, the earliest 80% form the training set (5,020 bouts), and the most recent 20% form the test set (1,255 bouts; fights from approximately late 2021 onwards). This choice prevents *time leakage*: using future fight data to predict past fights would give an optimistic and unrealistic evaluation. Random splits are inappropriate here because the same fighter appears in many bouts; a random split would leak information about individual fighters' long-run averages into the training set.

3.3 Model 1: Logistic Regression (baseline)

Logistic regression models the log-odds of Red winning as a linear function of the (scaled, imputed) difference features:

$$\log \frac{P(\text{Red wins})}{1 - P(\text{Red wins})} = \beta_0 + \boldsymbol{\beta}^\top \Delta\mathbf{x}.$$

Strengths: interpretable coefficients, fast to fit, well-calibrated probabilities. **Weaknesses**: assumes a linear decision boundary; cannot capture non-linear interactions between features. In our setting the feature space is noisy and low-signal; a linear boundary is a reasonable starting assumption. No regularisation is applied; coefficients are estimated by maximum likelihood.

3.4 Model 2: Elastic Net Logistic Regression (penalised, tuned)

Elastic Net adds a combined ℓ_1-ℓ_2 penalty to the log-likelihood:

$$\mathcal{L}(\boldsymbol{\beta}) - \lambda \left[\frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 + \alpha \|\boldsymbol{\beta}\|_1 \right],$$

where $C = 1/\lambda$ controls regularisation strength and α (called `l1_ratio` in scikit-learn) interpolates between Ridge ($\alpha = 0$) and Lasso ($\alpha = 1$). Elastic Net is preferable to pure Lasso when predictors are correlated: Lasso tends to arbitrarily discard one of a correlated group, whereas Ridge-like shrinkage stabilises the selection.

Hyperparameter tuning uses a nested cross-validation structure:

- **Outer split**: the chronological 80/20 holdout (shared with all models), providing an honest test estimate.
- **Inner CV**: 5-fold stratified cross-validation over the training set, optimising negative log loss.
- **Grid**: $C \in \{0.01, 0.1, 1.0, 10.0\}$, $\alpha \in \{0.1, 0.5, 0.9, 1.0\}$ (16 combinations).

The best parameters found were $C = 0.01$ (strong regularisation), $\alpha = 0.1$ (mostly Ridge with a light Lasso component). 21 out of 26 features survived the Lasso component with non-zero coefficients.

3.5 Model 3: Random Forest

A Random Forest trains $B = 400$ decision trees on bootstrap samples and averages their probability estimates. Each tree considers $\sqrt{p}$ randomly selected features at each split (bagging combined with the random subspace method). **Strengths**: captures non-linear structure and feature interactions without explicit specification; robust to outliers; provides feature importances via mean decrease in impurity. **Weaknesses**: less interpretable than linear models; can overfit with deep trees on small datasets. We set `min_samples_leaf = 10` to regularise tree depth and reduce variance; this was chosen by expert judgment given the dataset size (5,020 training observations, 26 features).

3.6 Model 4: Gradient Boosted Trees (XGBoost)

XGBoost builds trees sequentially, with each tree correcting the residuals of the current ensemble. The learning rate (`learning_rate` = 0.05) limits each tree's contribution to prevent overfitting, and stochastic subsampling (`subsample` = 0.8, `colsample_bytree` = 0.8) introduces variance reduction. The minimum child weight (`min_child_weight` = 5) acts as additional regularisation on leaf size. **Strengths**: strong empirical performance; handles missing values natively; flexible regularisation. **Weaknesses**: more hyperparameters to configure; prone to overfitting without careful tuning; less interpretable than logistic regression.

3.7 Evaluation metrics

All models are evaluated on the held-out chronological test set and compared against the bookmaker-implied probability baseline. We report:

- **Accuracy**: fraction of correctly predicted outcomes (threshold at 0.5).
- **Log loss** (cross-entropy): measures calibration and sharpness of probability estimates; lower is better. Treated as the primary ranking metric because it rewards well-calibrated probabilities, not just the sign of the prediction.
- **Brier score**: mean squared error between predicted probabilities and binary outcomes; lower is better.
- **ROC AUC**: area under the receiver operating characteristic curve; measures discrimination ability independently of the decision threshold.

4 Results and Interpretation

4.1 Model performance

Table 2 compares all models on the 1,255-bout test set. The bookmaker benchmark is shown for reference; Δ values are signed differences relative to the bookmaker (negative Δ log loss / Brier is better; positive Δ accuracy / AUC is better).

Table 2: Test-set evaluation results (chronological holdout, $n_{\text{test}} = 1{,}255$).

Model	Acc	Δ Acc	Log Loss	Δ LL	Brier	ROC AUC
Bookmaker (baseline)	0.696	—	0.593	—	0.204	0.743
Naive (majority class)	0.571	−0.124	0.693	+0.100	0.250	0.500
Logistic Regression	0.635	−0.060	0.636	+0.043	0.223	0.673
Elastic Net (nested CV)	0.640	−0.056	0.635	+0.041	0.222	0.682
Random Forest	**0.636**	−0.060	**0.633**	+0.040	**0.221**	**0.693**
XGBoost	0.632	−0.064	0.636	+0.043	0.223	0.670

Random Forest achieves the best log loss (0.633), Brier score (0.221), and ROC AUC (0.693). However, the differences among Logistic Regression, Elastic Net, and Random Forest are small (log loss range < 0.004), suggesting that the dominant bottleneck is the limited predictive content of public statistics, not the model family. XGBoost performs slightly worse despite its higher complexity, consistent with a higher-variance model that cannot be offset by the available signal.

All models comfortably beat the naive majority-class baseline (AUC 0.5) but remain well below the bookmaker (AUC 0.743). The persistent $\approx$5.5 percentage-point accuracy gap to the bookmaker quantifies the information advantage of market prices over publicly available fight statistics.

4.2 Confusion matrices and class-level behaviour

Table 3 shows confusion matrix entries and precision/recall for each model. A striking pattern is that all models predict Red wins the large majority of the time, driven by the 58.1% class imbalance. Recall for the Red class (fraction of actual Red wins correctly called) is above 0.83 for all linear/ensemble models, while recall for the Blue class is much lower. XGBoost achieves the most balanced predictions (TN = 233, highest among all models), reflecting the effect of `min_child_weight` regularisation.

Table 3: Confusion matrix entries and precision/recall on the test set (n = 1,255; 717 Red wins, 538 Blue wins). Precision and recall are reported for the Red-wins class (positive class).

Model	TN	FP	FN	TP	Precision	Recall
Logistic Regression	202	336	122	595	0.639	0.830
Elastic Net (nested CV)	198	340	112	605	0.640	0.844
Random Forest	201	337	120	597	0.639	0.833
XGBoost	233	305	157	560	0.647	0.781

4.3 ROC curves

Figure 5 shows the ROC curve for the best-performing model (Random Forest) alongside the bookmaker benchmark. The bookmaker ROC curve lies clearly above the model curve across all thresholds, confirming that market odds encode more discriminative information than our feature set.

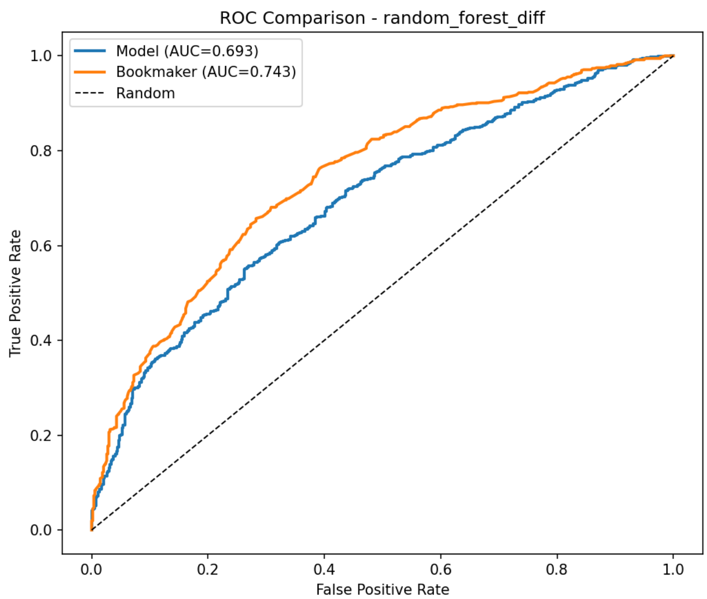

Figure 5: ROC curves for the Random Forest model versus the bookmaker baseline on the test set. The bookmaker (AUC = 0.743) substantially outperforms the best statistical model (AUC = 0.693).

4.4 Coefficient interpretation (Elastic Net)

Table 4 shows the largest-magnitude Elastic Net coefficients after nested-CV tuning ($C = 0.01$, $\alpha = 0.1$). The strong regularisation ($C = 0.01$) shrinks all coefficients toward zero; 21 out of 26 features survive with non-zero weights. Positive coefficients indicate features that, when larger for the Red corner, increase the predicted probability of a Red victory.

Table 4: Top 8 Elastic Net coefficients (sorted by $|\hat{\beta}|$). The Lasso component zeroed out 5 features; the remaining 21 are listed in the full CSV at `reports/metrics/elasticnet_nested_cv_diff_coefficients.csv`.

Feature	$\hat{\beta}$	Interpretation
`AgeDiff`	-0.259	Older Red fighter $\rightarrow$ lower Red win probability
`AvgTDLandedDiff`	$+0.123$	Red takedown advantage $\rightarrow$ higher Red win prob.
`ReachCmsDiff`	$+0.105$	Red reach advantage $\rightarrow$ higher Red win prob.
`CurrentWinStreakDiff`	$+0.104$	Longer Red active win streak $\rightarrow$ higher Red win prob.
`LossesDiff`	-0.087	More career losses for Red $\rightarrow$ lower Red win prob.
`WeightLbsDiff`	$+0.080$	Red weight advantage $\rightarrow$ higher Red win prob.
`LongestWinStreakDiff`	$+0.069$	Longer Red peak win streak $\rightarrow$ higher Red win prob.
`DrawsDiff`	$+0.057$	More draws for Red $\rightarrow$ slight Red win advantage

The negative age coefficient warrants attention: `AgeDiff` $= \text{Age}_{\text{Red}} - \text{Age}_{\text{Blue}}$, so the negative sign means older Red fighters are predicted to win less often. Since Red fighters tend to be older (mean `AgeDiff` $= 0.56$ years), this coefficient captures a pattern where relatively younger opponents outperform statistical expectations, possibly reflecting the age-related decline of veteran Red-corner fighters.

4.5 Feature importance (Random Forest)

Figure 6 shows the top-20 feature importances from the Random Forest, measured by mean decrease in impurity. The top three features are `AgeDiff`, `AvgSigStrLandedDiff`, and `AvgTDLandedDiff`. This broadly agrees with the Elastic Net coefficient magnitudes (Table 4), giving confidence that these features carry genuine signal rather than being artefacts of one particular model class.

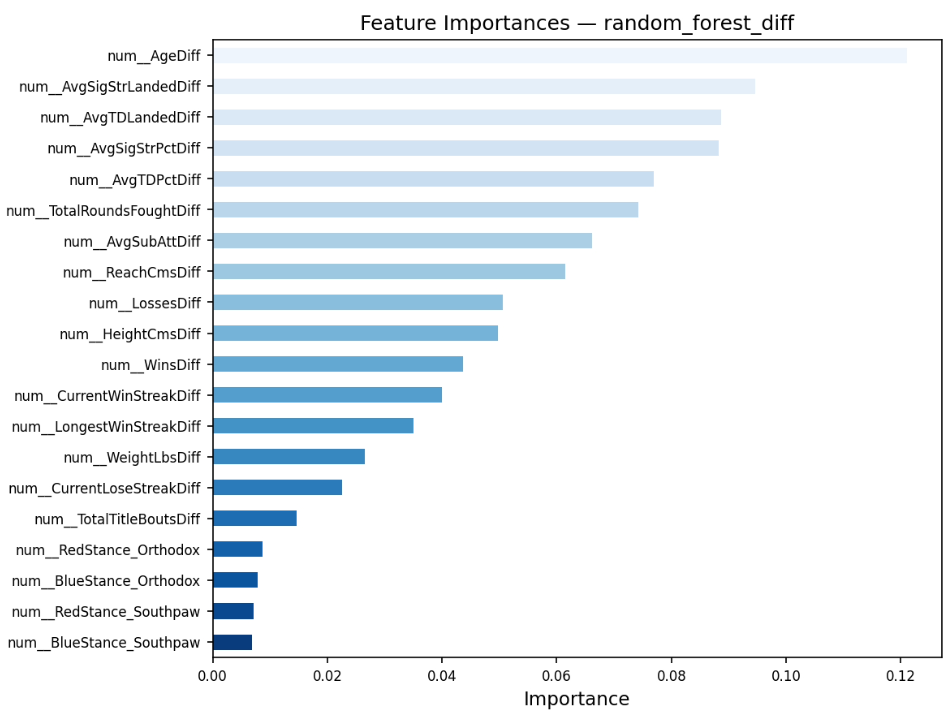

Figure 6: Random Forest feature importances (mean decrease in impurity) for the top 20 features. `AgeDiff` dominates; grappling and striking averages follow.

4.6 Bias-variance discussion

Logistic regression has the lowest variance but a linear decision boundary that cannot capture threshold effects (e.g., a reach advantage may matter very differently in the heavyweight division versus flyweight). Random Forest and XGBoost have lower bias and can fit interactions, but their higher variance is partially controlled through regularisation (`min_samples_leaf = 10` and `min_child_weight = 5`). Despite the added flexibility, ensemble methods do not substantially improve over regularised logistic regression (log loss difference < 0.004). This suggests that the signal-to-noise ratio in public UFC statistics is intrinsically low, and the variance cost of flexible models nearly cancels their bias benefit.

The fact that $C = 0.01$ (strong regularisation) was selected by inner CV for the Elastic Net also points in this direction: the CV is protecting against overfitting to noise, not trying to fit a complex signal.

4.7 Limitations

- **Bookmaker gap**: The persistent $\approx$5.5 percentage-point accuracy gap to the bookmaker suggests important predictors (training camp quality, recent injuries, travel, stylistic matchups) are unobservable from public statistics.
- **Missing data**: Many earlier bouts lack complete fighter statistics; median imputation introduces noise that may weaken model discrimination for these fights.
- **Label asymmetry**: The Red/Blue assignment is non-random; Red fighters tend to be higher-ranked and older. This means the difference features carry a structural bias toward the Red corner that the model must implicitly correct for.
- **Fixed hyperparameters**: Hyperparameters for Random Forest and XGBoost were not formally tuned; nested CV for these models might close the small remaining gap to the bookmaker.
- **Chronological split**: Using only the last 20% of fights as the test set means test performance is measured on recent fights. If fighter statistics tracking or matchmaking strategy has changed over time, the train-test gap may reflect distributional shift rather than true generalisation error.

5 Summary

We framed UFC fight outcome prediction as a binary classification problem and compared three model families on a chronological test set of 1,255 bouts. All models comfortably beat the naive majority-class baseline (AUC 0.5) and extract meaningful signal from the difference feature representation. No model closes the gap to the bookmaker benchmark (AUC 0.743), confirming that market-implied probabilities encode information not accessible from public fighter statistics alone.

The Random Forest achieved the best overall performance (log loss 0.633, AUC 0.693), but the margin over regularised logistic regression is small. The Elastic Net with nested cross-validation provided the best interpretability: inner cross-validation selected strong regularisation ($C = 0.01$), and 21 out of 26 features survived the Lasso step. The most important predictors across both model families were average takedowns landed, age difference, reach advantage, and win-streak difference.

The most practically significant finding is the consistency of the bookmaker gap across all models. It establishes a clear ceiling: the information contained in publicly available cumulative career statistics is real but insufficient, and closing the gap would require access to information that is currently not captured in any public dataset.